

Practice Questions for 2024-10-01-DualGeneralLP

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1 Questions

1. Given the primal problem (P) $\max \quad \zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$, and its dual problem (D) $\min \quad \xi = b^T y$ subject to $A^T y \geq c$ and $y \geq 0$, as shown in Slide 1, what is the expression for the Lagrangian function $L(x, y)$ used in the derivation of the dual problem?
 - a. $L(x, y) = c^T x + y^T (Ax - b)$
 - b. $L(x, y) = c^T x - y^T (Ax - b)$
 - c. $L(x, y) = c^T x + y^T (b - Ax)$
 - d. $L(x, y) = b^T y - x^T (A^T y - c)$
 - e. $L(x, y) = b^T y + x^T (A^T y - c)$
2. Explain the significance of introducing the variables x^+ and x^- in Slide 2 when transforming the general linear program (MI).
 - a. To ensure the objective function remains linear.
 - b. To handle unbounded variables and guarantee the existence of a solution.
 - c. To simplify the constraint matrix A and make the problem easier to solve.
 - d. To convert inequality constraints into equality constraints.
 - e. To allow the use of the simplex method directly without further transformations.
3. In Slide 3, the derivation leads to the condition $A^T y + z \geq c$ and $A^T y + z \leq c$. Explain why this ultimately simplifies to $A^T y + z = c$ in the dual problem (D).
 - a. It's a mathematical simplification to reduce the number of constraints.
 - b. It's a consequence of the non-negativity constraints on x^+ and x^- .
 - c. It's an assumption made to ensure the dual problem is feasible.
 - d. It's a result of minimizing the Lagrangian function with respect to x .
 - e. It's necessary to ensure strong duality holds between the primal and dual problems.
4. In Slide 6, the minimization of the Lagrangian leads to a piecewise function. Explain the significance of the condition $c^T - Y^T A - Z^T = 0$ and its implication for the dual problem.
 - a. It ensures the dual problem is always feasible.

- b. It guarantees the existence of a finite optimal solution for the dual problem.
 - c. It simplifies the calculation of the dual objective function.
 - d. It is a necessary condition for weak duality to hold.
 - e. It is a sufficient condition for strong duality to hold.
5. Explain the significance of introducing the variables x^+ and x^- in Slide 2 when transforming the general linear program (M1).
- a. To ensure the objective function remains linear.
 - b. To handle unbounded variables in the primal problem.
 - c. To simplify the derivation of the dual problem.
 - d. To convert inequality constraints into equality constraints.
 - e. To guarantee the feasibility of the primal problem.
6. Given the general linear program (M1) in Slide 2, $\max c^T x$ subject to $Ax \leq b$ and $x \leq u$, what is the objective function of the transformed problem after introducing the variables x^+ and x^- ?
- a. $c^T x^+$
 - b. $c^T x^-$
 - c. $c^T (x^+ + x^-)$
 - d. $c^T (x^+ - x^-)$
 - e. $c^T (x^- - x^+)$
7. In Slide 3, the derivation of the dual problem involves the inequalities $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$. What condition is ultimately imposed to simplify the constraints of the dual problem?
- a. $y^T A + z^T \geq 0$
 - b. $y^T A + z^T \leq 0$
 - c. $A^T y + z \geq c$
 - d. $A^T y + z = c$
 - e. $y^T b + z^T u = 0$
8. Referring to Slide 4, what is the Lagrangian function, $L(x, y, z)$, for the linear program (M2) given by $\max c^T x$ subject to $Ax \leq b$ and $x \leq u$?
- a. $L(x, y, z) = c^T x + y^T (Ax - b) + z^T (x - u)$
 - b. $L(x, y, z) = c^T x - y^T (Ax - b) - z^T (x - u)$
 - c. $L(x, y, z) = c^T x + y^T (Ax + b) + z^T (x + u)$
 - d. $L(x, y, z) = c^T x - y^T (Ax + b) - z^T (x + u)$
 - e. $L(x, y, z) = y^T (Ax - b) + z^T (x - u)$

9. In Slide 6, the minimization of the Lagrangian leads to a piecewise function. What condition must be satisfied for the minimum value of the Lagrangian to be finite (not $+\infty$)?
- a. $c^T - Y^T A - Z^T > 0$
 - b. $c^T - Y^T A - Z^T < 0$
 - c. $c^T - Y^T A - Z^T = 0$
 - d. $Y^T b + Z^T u > 0$
 - e. $Y^T b + Z^T u < 0$
10. Given the general linear program (MI) in Slide 2, $\max \quad c^T x$ subject to $Ax \leq b$ and $x \leq u$, what is the objective function of the transformed problem after introducing the variables x^+ and x^- ?
- a. $c^T x^+$
 - b. $c^T x^-$
 - c. $c^T (x^+ + x^-)$
 - d. $c^T (x^+ - x^-)$
 - e. $c^T (x^- - x^+)$

2 Answers

1. Given the primal problem (P) $\max \quad \zeta = c^T x$ subject to $Ax \leq b$ and $x \geq 0$, and its dual problem (D) $\min \quad \xi = b^T y$ subject to $A^T y \geq c$ and $y \geq 0$, as shown in Slide 1, what is the expression for the Lagrangian function $L(x, y)$ used in the derivation of the dual problem?
 - a. This is incorrect. The signs are reversed in the constraint term. The Lagrangian should be minimized when the constraints are satisfied.
 - b. The correct expression for the Lagrangian function is $L(x, y) = c^T x - y^T (Ax - b)$. The Lagrangian incorporates the objective function and the constraints, with y as Lagrange multipliers for the inequality constraints $Ax \leq b$. The negative sign ensures that the Lagrangian is minimized when the constraints are satisfied.
 - c. This is incorrect. The signs are reversed in the constraint term. The Lagrangian should be minimized when the constraints are satisfied.
 - d. This is incorrect. This expression represents a Lagrangian related to the dual problem, not the primal problem.
 - e. This is incorrect. This expression represents a Lagrangian related to the dual problem, not the primal problem, and the signs are reversed.
2. Explain the significance of introducing the variables x^+ and x^- in Slide 2 when transforming the general linear program (MI).
 - a. While the transformation maintains linearity of the objective function, it's not the primary significance. The main purpose is to handle variables without inherent non-negativity constraints.
 - b. The introduction of $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$ allows us to represent any real-valued variable x as the difference of two non-negative variables ($x = x^+ - x^-$). This is crucial because the standard form of a linear program requires non-negativity constraints. Without this transformation, the duality theory presented in the subsequent slides would not directly apply to the general linear program (MI) which does not have the non-negativity constraint on x .
 - c. The transformation doesn't directly simplify the constraint matrix A . The focus is on the variables, not the matrix itself.
 - d. The transformation doesn't directly convert inequality constraints to equality constraints. That happens later in the derivation of the dual.
 - e. The transformation is a necessary step before applying duality theory and the simplex method, not a direct application of the simplex method itself.
3. In Slide 3, the derivation leads to the condition $A^T y + z \geq c$ and $A^T y + z \leq c$. Explain why this ultimately simplifies to $A^T y + z = c$ in the dual problem (D).
 - a. While it simplifies the constraints, the simplification is a direct consequence of the minimization process, not an arbitrary reduction.
 - b. The non-negativity constraints are important, but the equality arises from the minimization of the Lagrangian, not directly from the non-negativity of x^+ and x^- .
 - c. It's not an assumption; it's a direct consequence of the mathematical derivation and the minimization of the Lagrangian.
 - d. The derivation in Slide 3 involves minimizing the Lagrangian function. The condition $A^T y + z \geq c$ arises from the coefficient of x^+ and $A^T y + z \leq c$ from the coefficient of x^- . To minimize the

Lagrangian, the coefficient of x must be zero, which implies $c^T - Y^T A - Z^T = 0$, leading to the equality constraint $A^T y + z = c$ in the dual problem. If the coefficient were non-zero, the Lagrangian would be unbounded.

- e. While strong duality is related, the equality constraint is a direct result of the minimization process, not a requirement for strong duality to hold.
4. In Slide 6, the minimization of the Lagrangian leads to a piecewise function. Explain the significance of the condition $c^T - Y^T A - Z^T = 0$ and its implication for the dual problem.
 - a. Feasibility is related, but the condition primarily ensures a finite optimal solution, not necessarily feasibility.
 - b. If $c^T - Y^T A - Z^T \neq 0$, then the Lagrangian is unbounded below, meaning the dual problem is unbounded. Therefore, the condition $c^T - Y^T A - Z^T = 0$ is crucial for the dual problem to have a finite optimal solution. This condition directly leads to the constraints of the dual problem, ensuring that the dual objective function is finite and the dual problem is well-defined.
 - c. While it simplifies the objective function, the main significance is in ensuring a finite optimal solution.
 - d. Weak duality holds regardless of this condition; this condition is about the boundedness of the dual problem.
 - e. This is a necessary condition for strong duality, but not sufficient on its own. Feasibility of both primal and dual problems is also required.
 5. Explain the significance of introducing the variables x^+ and x^- in Slide 2 when transforming the general linear program (MI).
 - a. While the transformation maintains linearity in the objective function, it's not the primary reason for introducing x^+ and x^- . The main purpose is to handle potentially negative values of x .
 - b. The introduction of x^+ and x^- allows us to represent any real-valued variable x as the difference of two non-negative variables. This is crucial because the standard form of a linear program requires non-negative variables, and the original problem (MI) does not necessarily have this constraint. By expressing $x = x^+ - x^-$, where $x^+, x^- \geq 0$, we can handle variables that might take on negative values in the original formulation.
 - c. The transformation simplifies the derivation, but the core reason is to handle the absence of non-negativity constraints in the original problem.
 - d. Slack variables are used to convert inequalities to equalities, but x^+ and x^- are used to handle the sign of the variables.
 - e. The transformation doesn't guarantee feasibility; it allows us to work with a problem that can be more easily analyzed using duality theory.
 6. Given the general linear program (MI) in Slide 2, $\max \quad c^T x$ subject to $Ax \leq b$ and $x \leq u$, what is the objective function of the transformed problem after introducing the variables x^+ and x^- ?
 - a. This only considers the positive part of x .
 - b. This only considers the negative part of x .
 - c. This uses the sum instead of the difference.

- d. The transformation in Slide 2 replaces x with $x^+ - x^-$. Therefore, the objective function becomes $c^T(x^+ - x^-)$.
- e. This uses the opposite difference.
7. In Slide 3, the derivation of the dual problem involves the inequalities $(y^T A + z^T)x^+ + (-y^T A - z^T)x^- \leq y^T b + z^T u$. What condition is ultimately imposed to simplify the constraints of the dual problem?
- This is a condition on the coefficients, not the final constraint.
 - This is a condition on the coefficients, not the final constraint.
 - This is one of the intermediate inequalities, not the final simplification.
 - The slide shows that the condition $A^T y + z \leq c$ and $A^T y + z \geq c$ are combined to yield $A^T y + z = c$, simplifying the dual problem's constraints.
 - This is a condition on the objective function, not the constraints.
8. Referring to Slide 4, what is the Lagrangian function, $L(x, y, z)$, for the linear program (M2) given by $\max c^T x$ subject to $Ax \leq b$ and $x \leq u$?
- Incorrect signs for the constraints in the Lagrangian.
 - The Lagrangian function correctly incorporates the objective function and constraints with Lagrange multipliers y and z .
 - Incorrect signs for the constraints and b and u .
 - Incorrect signs for the constraints and b and u .
 - Missing the objective function.
9. In Slide 6, the minimization of the Lagrangian leads to a piecewise function. What condition must be satisfied for the minimum value of the Lagrangian to be finite (not $+\infty$)?
- This condition leads to an unbounded solution.
 - This condition leads to an unbounded solution.
 - Only when $c^T - Y^T A - Z^T = 0$ does the minimum value become $Y^T b + Z^T u$; otherwise, it's unbounded.
 - This is a condition on the objective function value, not the constraint for finiteness.
 - This is a condition on the objective function value, not the constraint for finiteness.
10. Given the general linear program (MI) in Slide 2, $\max c^T x$ subject to $Ax \leq b$ and $x \leq u$, what is the objective function of the transformed problem after introducing the variables x^+ and x^- ?
- Answer A is incorrect. This only considers the positive part of x , ignoring the negative part.
 - Answer B is incorrect. This only considers the negative part of x , ignoring the positive part.
 - Answer C is incorrect. This sums the positive and negative parts, which is not the correct transformation.
 - Answer D is correct. The transformation in Slide 2 replaces x with $x^+ - x^-$, directly substituting this into the original objective function $c^T x$ yields $c^T(x^+ - x^-)$.

- e. Answer E is incorrect. This subtracts the positive part from the negative part, reversing the correct transformation.